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(54) METHOD FOR ASSEMBLING BATTERY MODULE AND HOLDING MECHANISM

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(52) U.S. Cl.

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(57) ABSTRACT

A battery module assembling method includes a region forming step of forming a region where a cell stack is placeable between a pair of holding plates by applying a coercive deformation force that deforms at least one of the pair of holding plates against an elastic biasing force, a arranging step of arranging the cell stack in the region formed in the region forming step, and a pressing step of pressing the cell stack in the stacking direction by the elastic biasing force, by removing the coercive deformation force in a state where the cell stack is placed in the region.

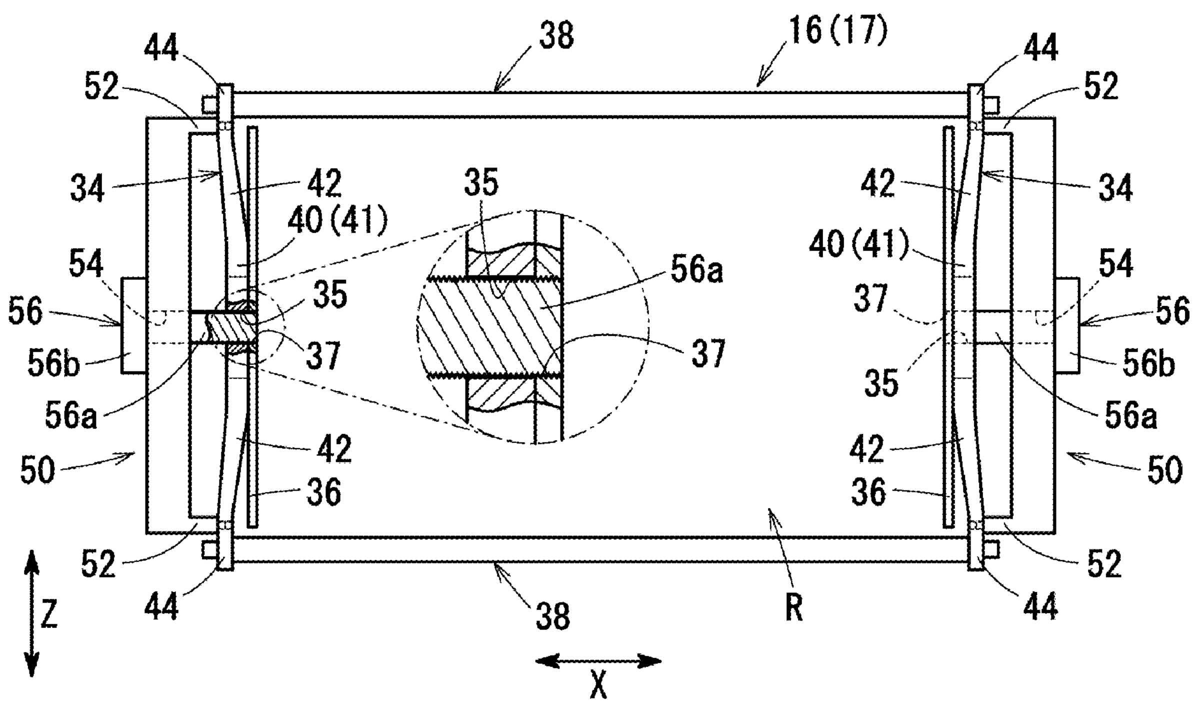


FIG. 1

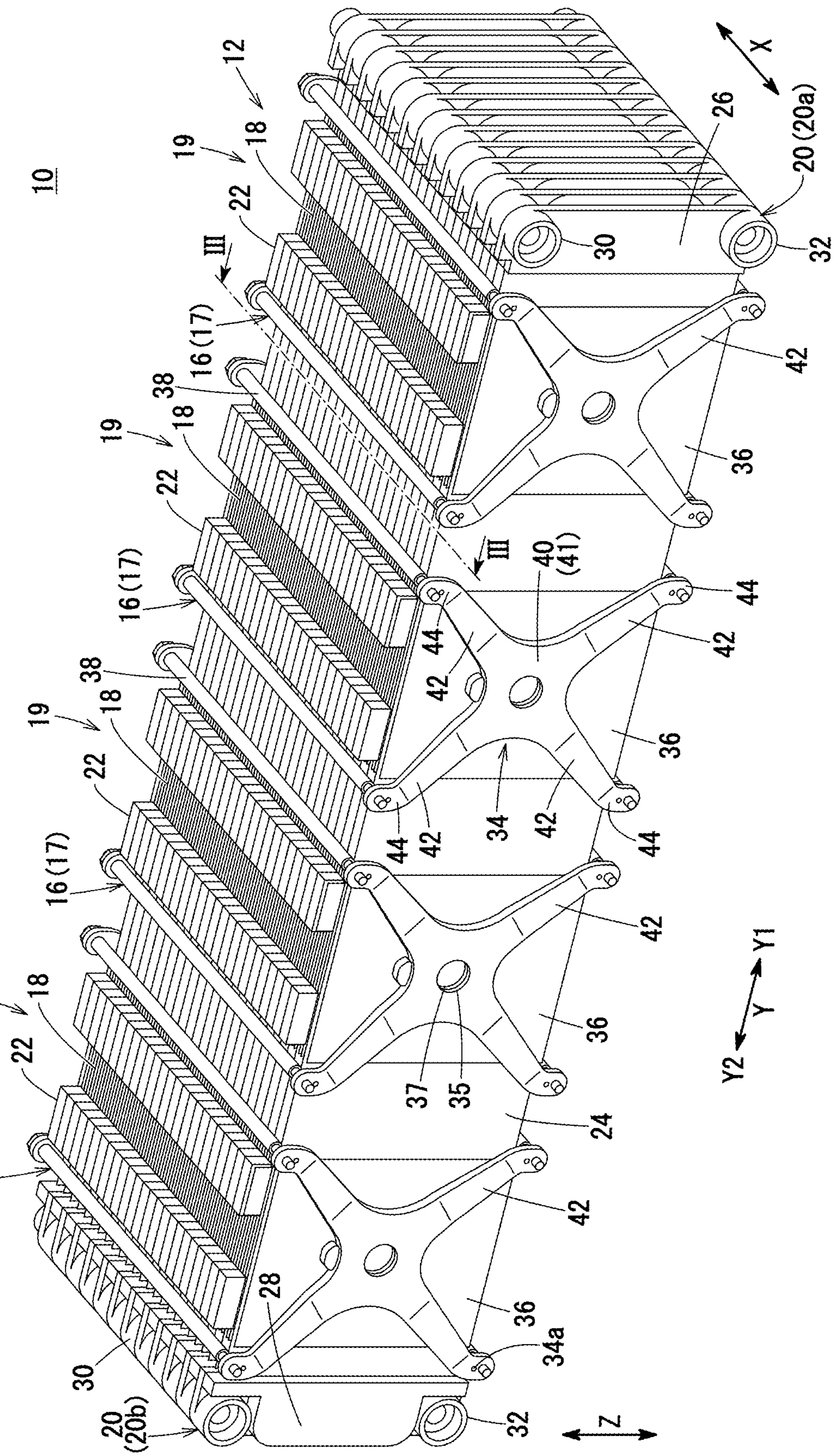

10
12
X
Y
Y1
Y2
Z
16(17)
18
19
20(20a)
20(20b)
22
24
26
28
30
32
34
34a
35
36
37
38
40(41)
42
44
III

FIG. 2

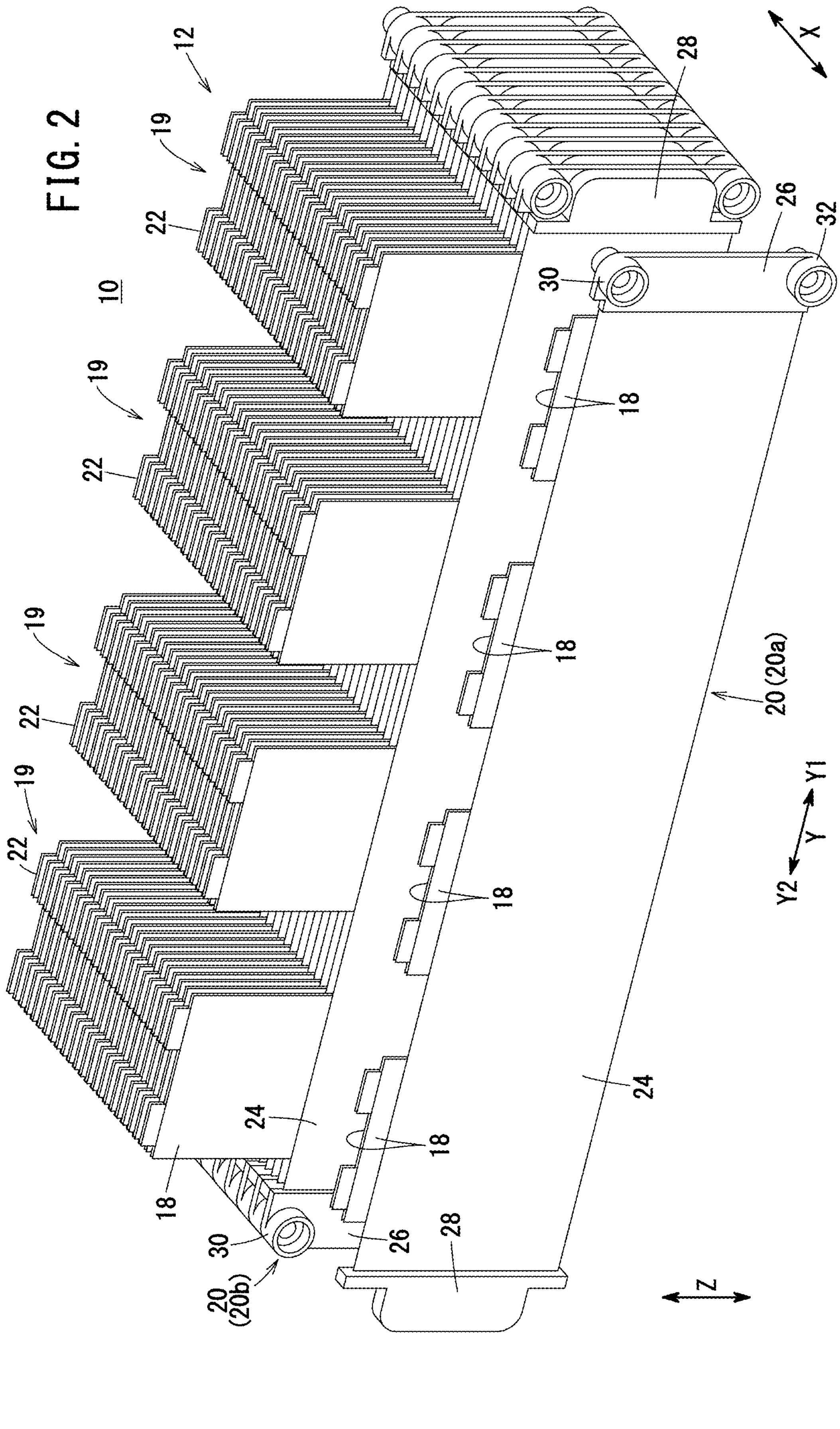


FIG. 3

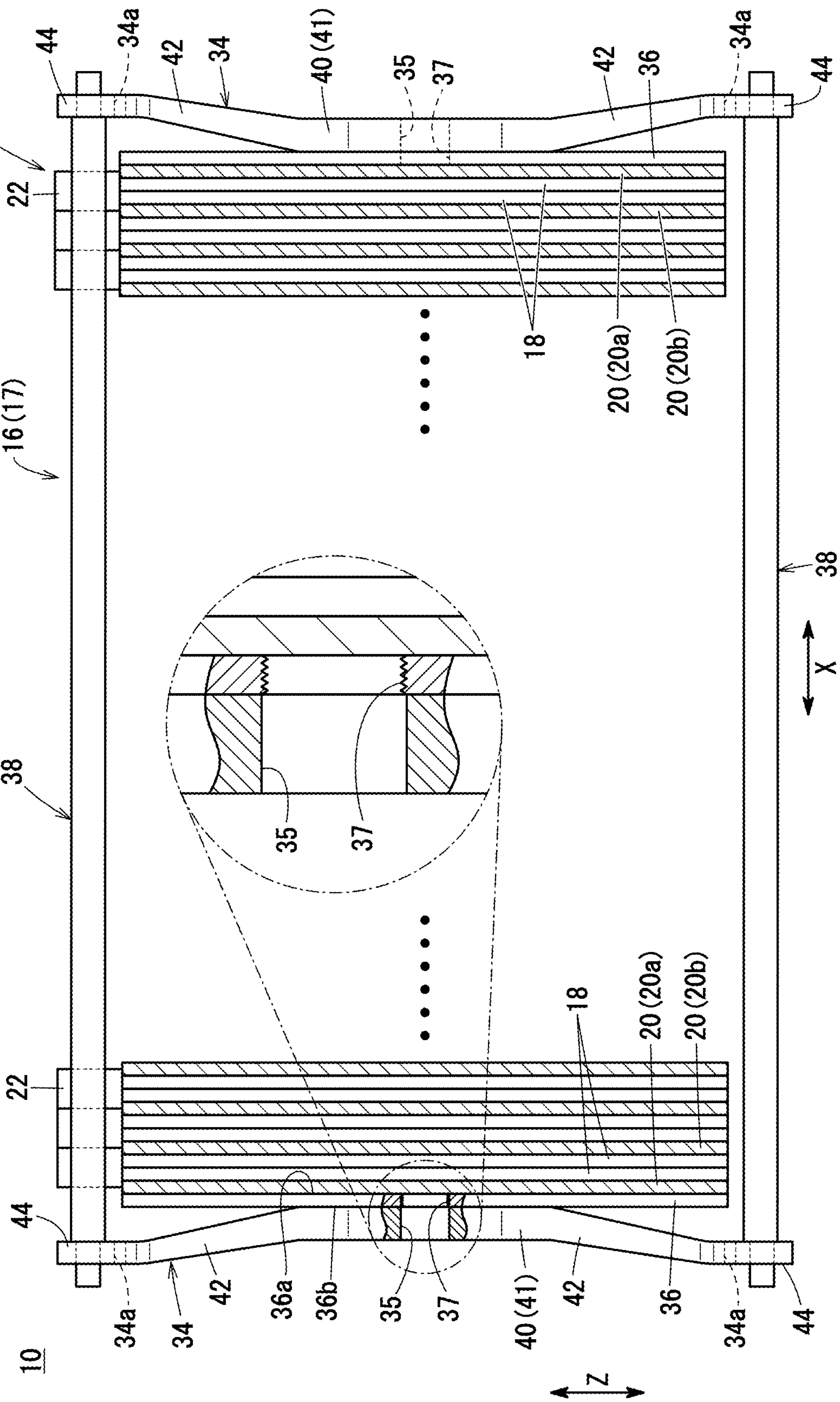
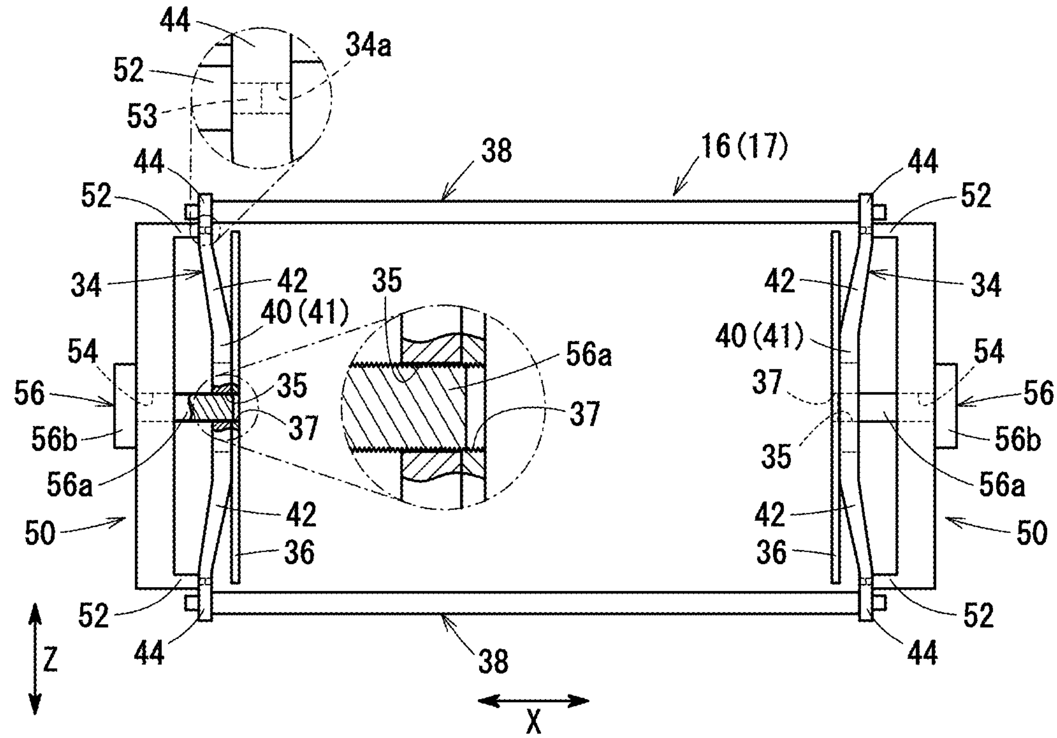

10
12
16 (17)
18
20 (20a)
20 (20b)
22
34
34a
35
36
36a
36b
37
38
40 (41)
42
44
X
Z

FIG. 4A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34a
52
53
44
38
16(17)
44
52
52
34
42
40(41)
35
42
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40(41)
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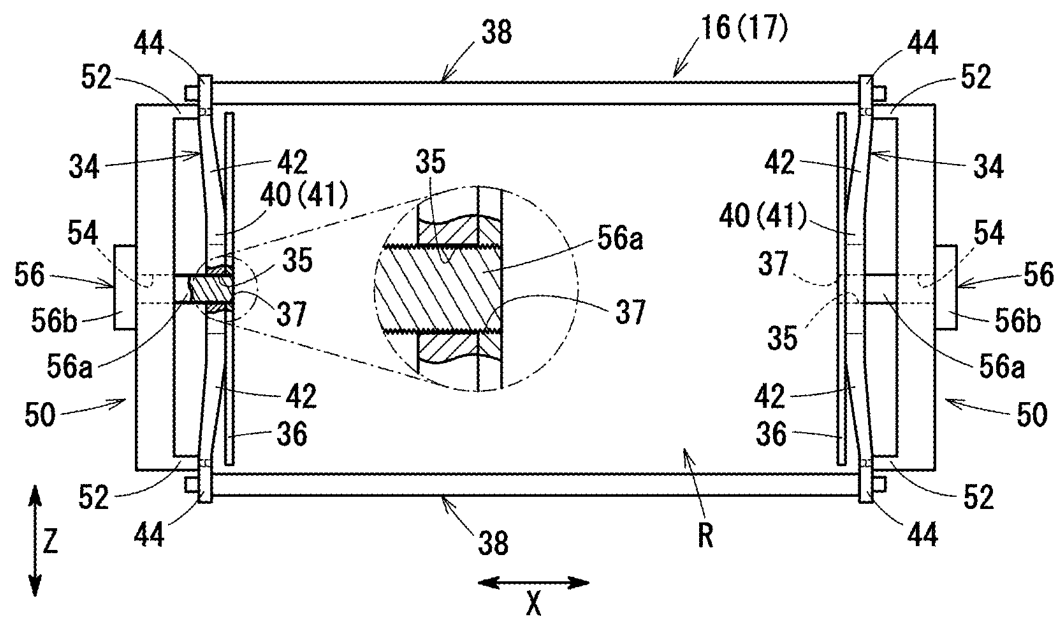
54
56
56b
56a
35
37
37
54
37
56
56b
35
56a
50
42
36
42
36
50
52
44
38
52
44
Z
X

FIG. 4B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38
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44
52
52
34
42
40(41)
35
42
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40(41)
54
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56b
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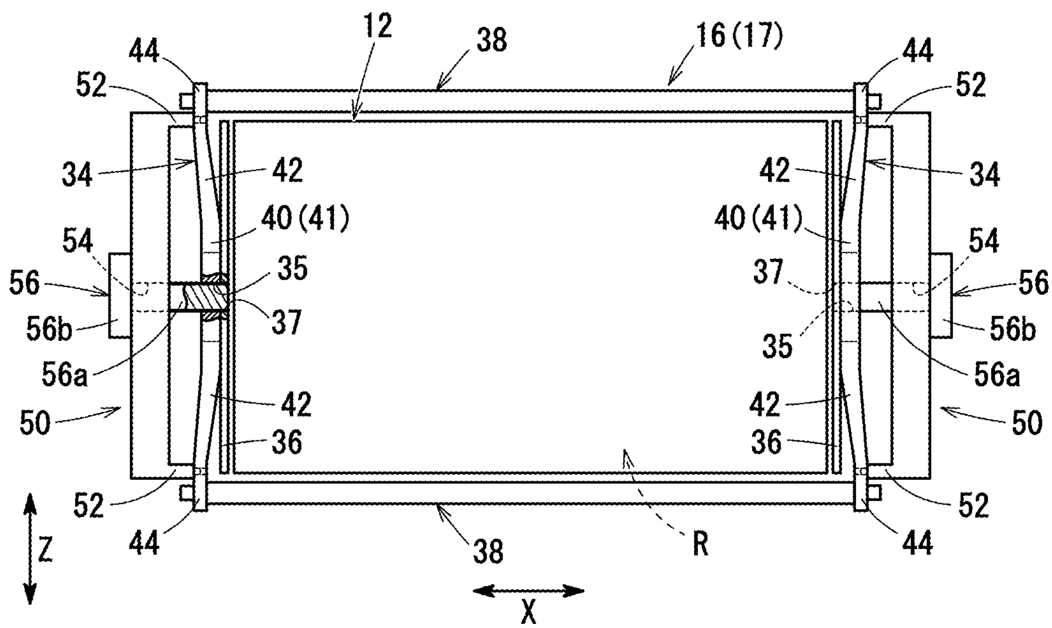
37
56a
37
54
37
56
35
56b
56a
50
42
36
42
36
50
R
52
44
38
52
44
Z
X

FIG. 5A


12
38
16 (17)
44
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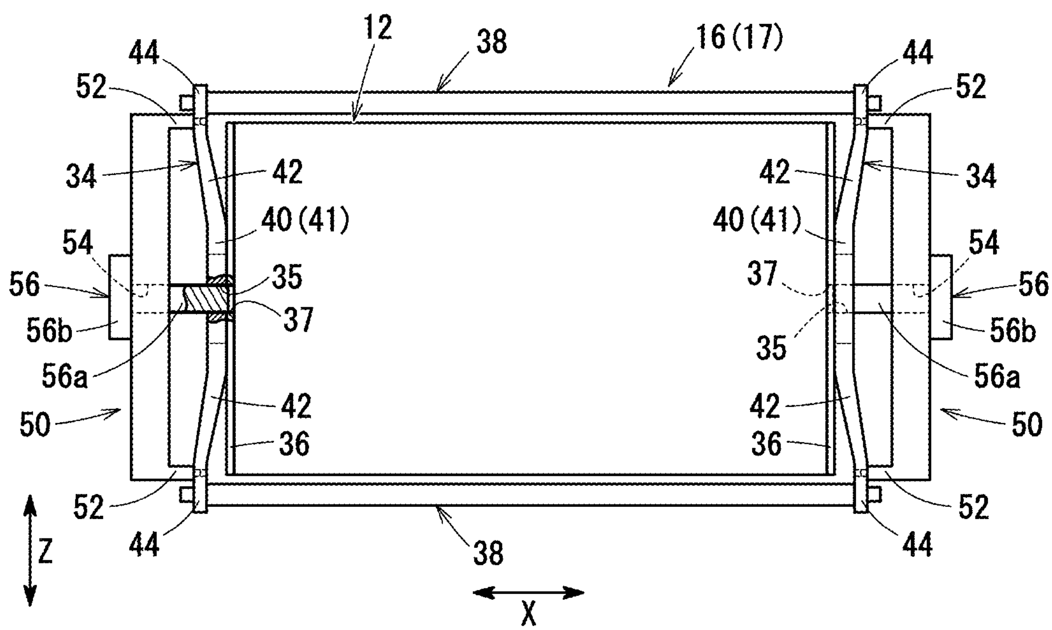
34
42
40 (41)
54
56
35
37
56b
56a
50
36
R
Z
X

FIG. 5B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16 (17)
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56a
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X

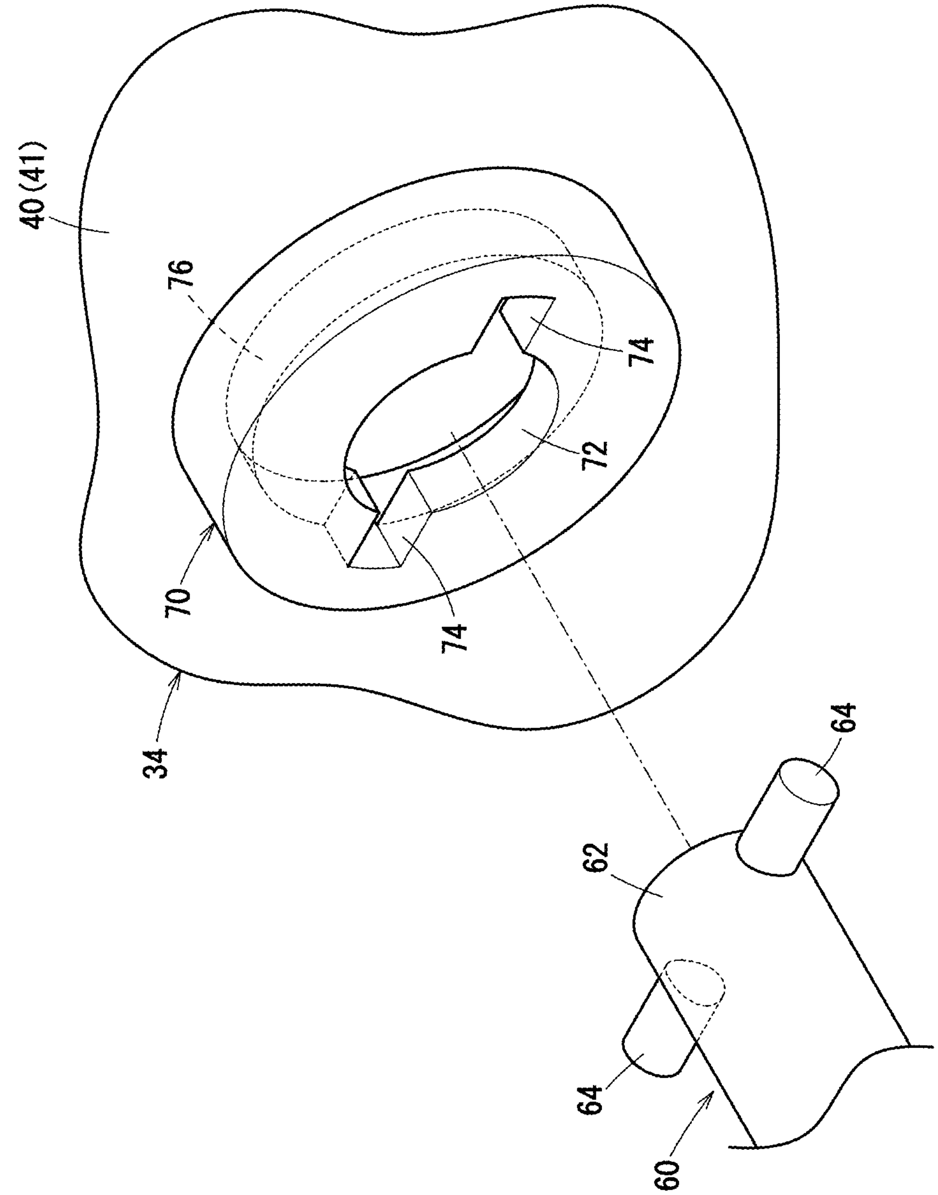
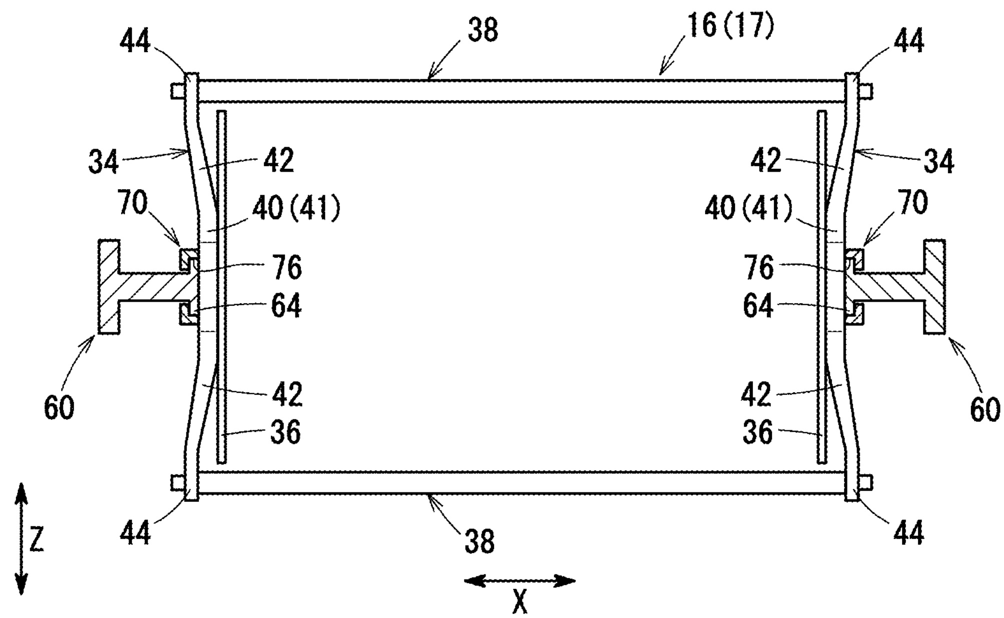


FIG. 6

FIG. 7A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34
42
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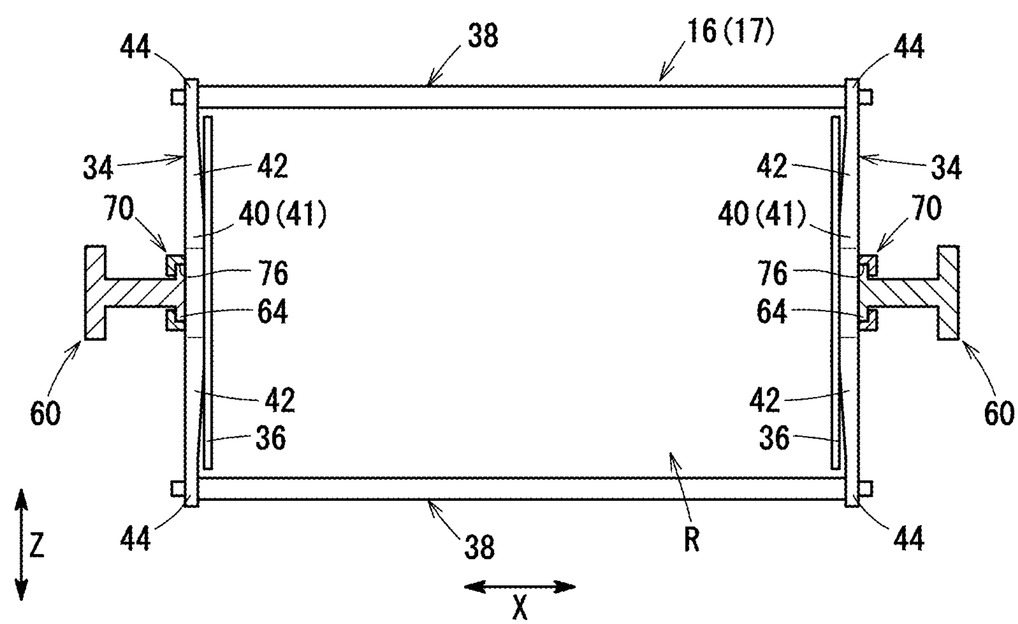
40(41)
76
64
60
36
Z
X

FIG. 7B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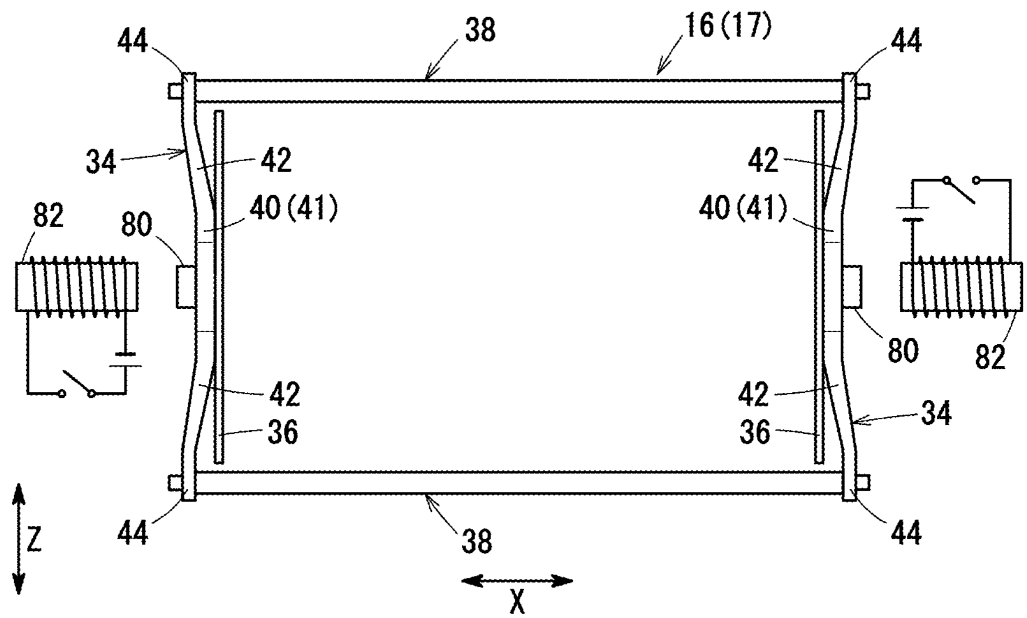
44
34
42
70
40(41)
76
64
60
36
R
Z
X

FIG. 8A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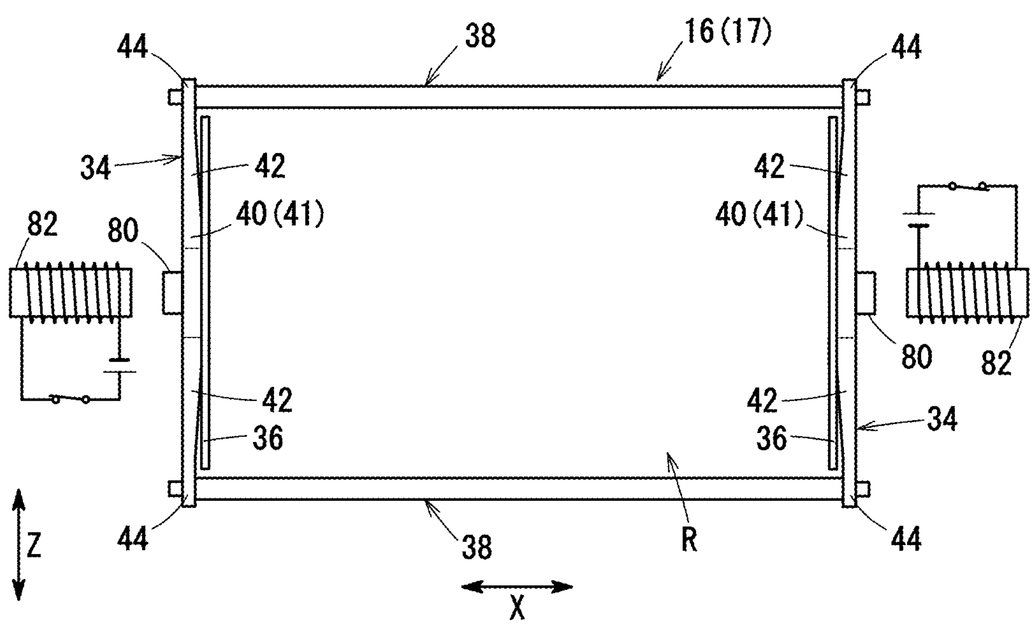
44
44
34
42
42
40(41)
40(41)
82
80
80
82
42
42
36
36
34
44
44
38
Z
X

FIG. 8B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40(41)
40(41)
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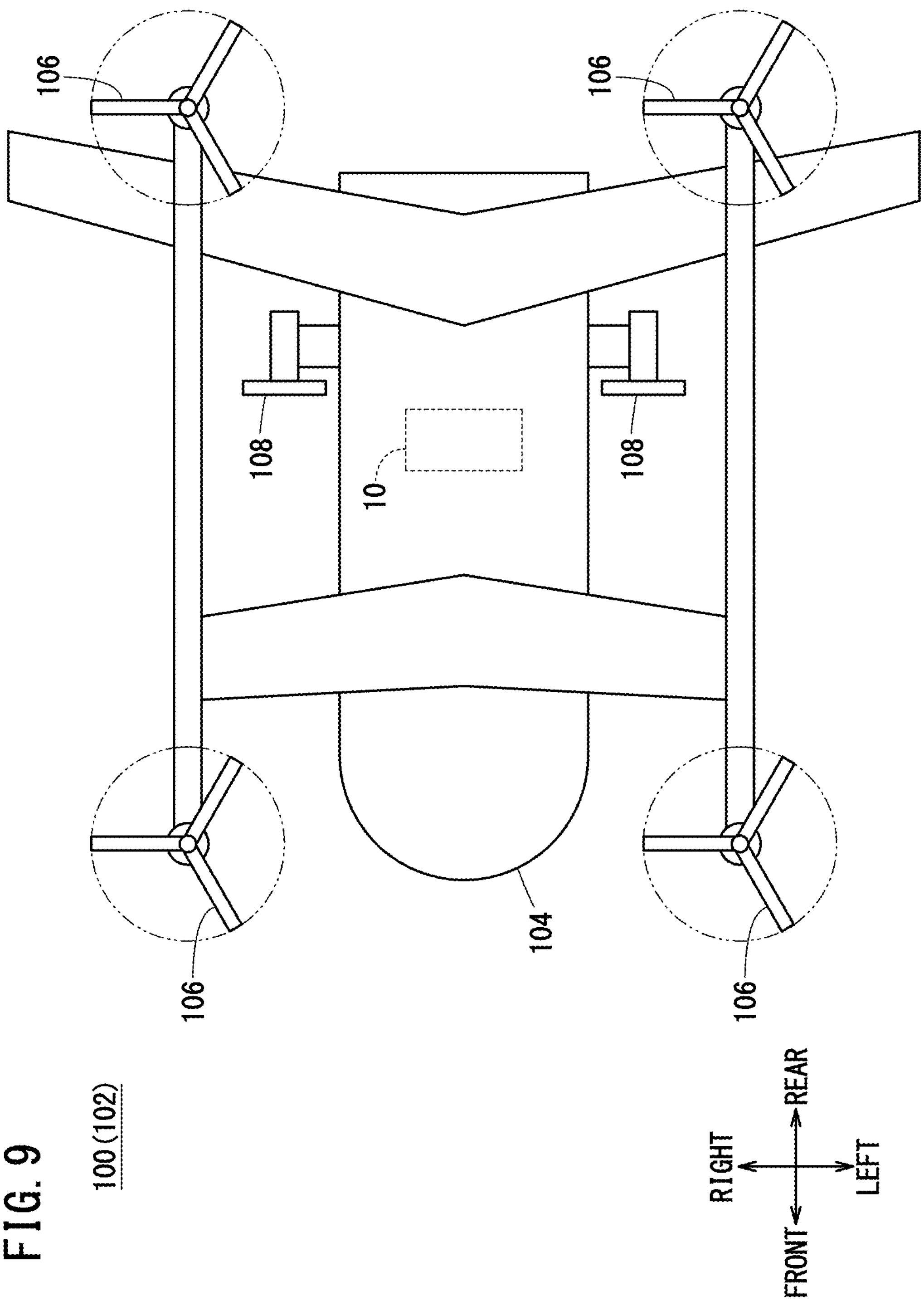
44
38
R
Z
X

FIG. 9



METHOD FOR ASSEMBLING BATTERY MODULE AND HOLDING MECHANISM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-034518 filed on Mar. 7, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a method for assembling a battery module and a holding mechanism.

Description of the Related Art

[0003] JP 2023-101130 A discloses a battery module including a cell stack formed by stacking battery cells and heat exchangers. The battery module further includes a battery frame as a holding mechanism that holds the cell stack by applying a tightening load from both sides of the cell stack. The battery frame prevents movement of the battery cells and the heat exchangers.

[0004] The battery frame has a pair of spring plates and four rod-like connecting members for connecting the pair of spring plates with each other. The spring plate has four arm portions that project radially from a central portion. When the cell stack is assembled to the battery frame, the screw structures (bolts and nuts) provided at both ends of the connecting member are tightened, whereby a tightening force is applied to the cell stack to hold the cell stack.

SUMMARY OF THE INVENTION

[0005] In the case of JP 2023-101130 A, since a pair of holding plates are fixed to the connecting members by means of a plurality of screw structures, respectively, it is difficult to equalize the distance between the nuts at both ends of the connecting member among the plurality of connecting members. Therefore, it is difficult to apply the stress evenly to the plurality of connecting members.

[0006] The present disclosure aims to solve the aforementioned problems.

[0007] A first aspect of the present disclosure is a method for assembling a battery module, which includes: a cell stack that includes battery cells and heat exchangers stacked on the battery cells; and a holding mechanism that holds both ends in a stacking direction of the cell stack, the holding mechanism includes a pair of holding plates that are elastically deformable, and a plurality of connecting members that connect the pair of holding plates with each other, and the cell stack is pressed inward in the stacking direction by the elastic biasing force of the pair of holding plates, whereby the cell stack is held, the method including: a region forming step of forming a region where the cell stack is configured to be placed between the pair of holding plates by applying coercive deformation force that deforms at least one of the pair of holding plates against the elastic biasing force; an arranging step of arranging the cell stack in the region formed in the region forming step, and a pressing step of pressing the cell stack in the stacking direction by the elastic biasing force, by removing the coercive deformation force in a state where the cell stack is placed in the region.

[0008] A second aspect of the present disclosure is a holding mechanism including a pair of holding plates that are elastically deformable and face each other, and a plurality of connecting members that connect the pair of holding plates with each other, wherein the holding mechanism is configured to hold a holding object placed between the pair of holding plates by the elastic biasing force of the pair of holding plates, each of the plurality of holding plates includes a pressing portion, and a plurality of arm portions extending from the pressing portion toward the plurality of connecting members, the plurality of arm portions are connected to the plurality of connecting members, and the pressing portion is configured to be moved in a direction to increase a space between the pair of holding plates by external force from the outside of the pair of holding plates.

[0009] According to the present disclosure, by externally applying a coercive deformation force to the holding plate, it is possible to form a region where the cell stack can be placed. In addition, by removing the coercive deformation force, it is possible to hold the cell stack by the elastic biasing force of the holding plate. Thus, the holding plate does not need to be tightened by the screw structure, and the holding plate and the connecting member can be fixed in such a way that positions are not adjustable. Therefore, the stress is applied evenly to the plurality of connecting members. As a result, the tightening force can be properly applied to the cell stack.

[0010] The above and other objects features and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings in which a preferred embodiment of the present invention is shown by way of illustrative example.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a perspective view of a battery module;

[0012] FIG. **2** is an exploded perspective view of a cell stack;

[0013] FIG. **3** is a schematic cross-sectional view taken along a line III-III of FIG. **1**;

[0014] FIG. **4**A is an explanatory diagram of a preparatory step of a region forming step in a method for assembling a battery module according to the first embodiment; FIG. **4**B is an explanatory diagram of a moving step of the region forming step;

[0015] FIG. **5**A is an explanatory diagram of a placement process;

[0016] FIG. **5**B is an explanatory diagram of a pressing step;

[0017] FIG. **6** is a perspective view of an engaging structure;

[0018] FIG. **7**A is an explanatory diagram of an engaging step of a region forming step in a method for assembling a battery module according to the second embodiment; FIG. **7**B is an explanatory diagram of a moving step of the region forming step;

[0019] FIG. **8**A is an explanatory diagram of a preparatory step of a region forming step in a method for assembling a battery module according to the third embodiment; FIG. **8**B is an explanatory diagram of a moving step of the region forming step; and

[0020] FIG. **9** is a schematic diagram of an aircraft in which the battery module is mounted.

DETAILED DESCRIPTION OF THE INVENTION

[0021] As shown in FIG. **9**, a battery module **10** is mounted in, for example, an aircraft **102** as a mobile object **100**. The aircraft **102** is, for example, an electric vertical take-off and landing aircraft (eVTOL). The aircraft **102** includes a fuselage **104**, multiple (e.g., four) VTOL rotors **106**, and multiple (e.g., two) cruise rotors **108**.

[0022] The VTOL rotor **106** generates an upward thrust force with respect to the aircraft **102**. The cruise rotor **108** generates a horizontal thrust force with respect to the aircraft **102**. The battery module **10** is placed inside the fuselage **104**. The battery module **10** supplies power to an electric motor (not shown) that drives each of the VTOL rotors **106** and the cruise rotors **108**. The mobile object **100** may also be, for example, a vehicle, a ship, or the like. The battery module **10** is not limited to the example where the battery module **10** is mounted in the mobile object **100**.

[0023] As shown in FIG. **1**, the battery module **10** includes a cell stack **12** and a plurality of battery frames **16**.

[0024] As shown in FIG. **2**, the cell stack **12** includes a plurality of battery cells **18** and a plurality of heat exchangers **20**. A single cell row **19** is formed of a plurality of battery cells **18** arranged in the direction of the arrow X. In this embodiment, four cell rows **19** are arranged in the direction of the arrow Y. The number of cell rows **19** may be three or less, or four or more. Only one cell row **19** may be provided in the battery module **10**.

[0025] The battery cells **18** and the heat exchangers **20** are arranged (stacked) in the direction of the arrow X. In the following, the X direction is also referred to as the "stacking direction". In addition, the direction toward the center of the battery module **10** in the X direction is expressed as "inward in the stacking direction". The direction away from the center of the battery module **10** in the X direction is expressed as "outward in the stacking direction".

[0026] The battery cell **18** is a laminate type battery. The battery cell **18** is formed in a rectangular plate shape. A plurality of terminal portions **22** project from one side of the battery cell **18** in the direction of the arrow Z. The battery cells **18** are connected in series with each other via the terminal portions **22**. The terminal portions **22** are conceptually illustrated. Electrical connecting members (not shown) are bonded to the terminal portions **22**.

[0027] The heat exchangers **20** include a plurality of first heat exchangers **20***a* and a plurality of second heat exchangers **20***b*. As shown in FIG. **2**, each first heat exchanger **20***a* has a plate-like water jacket **24**, a water supply-drainage header **26**, and a turn header **28**. The water jacket **24** extends in the direction of the arrow Y. A flow path through which cooling water circulates is formed in the water jacket **24**. Although not shown in detail, this flow path has a forward flow path for letting cooling water flow from the water supply-drainage header **26** toward the turn header **28** and a return flow path for letting cooling water flow from the turn header **28** toward the water supply-drainage header **26**.

[0028] The water supply-drainage header **26** is provided at one end (**Y1** direction side) in the longitudinal direction (direction of the arrow Y) of the water jacket **24**. The water supply-drainage header **26** supplies cooling water to and discharges cooling water from the water jacket **24**.

[0029] The water supply-drainage header **26** has a water supply port **30** and a water drainage port **32**. The water supply port **30** supplies cooling water to the forward flow path of the water jacket **24**. The water supply ports **30** of the first heat exchangers **20***a* adjacent to each other are connected liquid-tightly to each other. The drainage port **32** discharges the cooling water from the return flow path of the water jacket **24**. The drainage ports **32** of the first heat exchangers **20***a* adjacent to each other are connected liquid-tightly to each other.

[0030] Although the details are not illustrated, the water supply ports **30** of the first heat exchangers **20***a* adjacent to each other are connected to be relatively movable in the X direction so that the expansion of the battery cells **18** in the X direction caused by heat generation or deterioration of the battery cells **18** can be absorbed. Similarly, the drainage ports **32** of the first heat exchangers **20***a* adjacent to each other are connected to each other to be relatively movable in the X direction.

[0031] The turn header **28** is provided at the other end (**Y2** direction side) of the water jacket **24** in the longitudinal direction. The turn header **28** receives cooling water from the forward flow path of the water jacket **24** and lets the cooling water flow to the return flow path of the water jacket **24**.

[0032] The second heat exchanger **20***b* has a water jacket **24**, a water supply-drainage header **26**, and a turn header **28**, as the first heat exchanger **20***a*. However, the second heat exchanger **20***b* is arranged in a different direction from the first heat exchanger **20***a* in the Y-direction. Therefore, in the case of the second heat exchanger **20***b*, the water supply-drainage header **26** is arranged on the **Y2** direction side of the water jacket **24**, and the turn header **28** is arranged on the **Y1** direction side of the water jacket **24**.

[0033] The first heat exchanger **20***a* and the second heat exchanger **20***b* are alternately arranged in the direction of the arrow X. Thus, the water supply-drainage header **26** of one of the first heat exchanger **20***a* and the second heat exchanger **20***b* and the turn header **28** of the other of the first heat exchanger **20***a* and the second heat exchanger **20***b* are adjacent to each other in the stacking direction (X direction).

[0034] As shown in FIG. **3**, two battery cells **18** are stacked in the direction of the arrow X between the first heat exchanger **20***a* and the second heat exchanger **20***b* that are adjacent to each other.

[0035] As shown in FIG. **1**, in this embodiment, four battery frames **16** are provided corresponding to the four cell rows **19**. The number of battery frames **16** may be three or less or five or more depending on the number of cell rows **19**.

[0036] As shown in FIGS. **1** and **3**, the battery frame **16** includes a pair of holding plates **34**, a pair of pressure receiving plates **36**, and a plurality of connecting members **38**. The pair of holding plates **34** are placed at the end portions of the battery module **10** in the direction of the arrow X. The battery frame **16** is a holding mechanism **17** that can hold, by the elastic biasing force of the pair of holding plates **34**, a holding object (cell stack **12**) placed between the pair of holding plates **34**.

[0037] The pair of holding plates **34** are located outward in the stacking direction of the battery cells **18**. The holding plate **34** is made of, for example, titanium alloy. The holding plate **34** may be made of a metal material other than titanium alloy.

[0038] As shown in FIG. **1**, the holding plate **34** is formed in an X-shape when viewed from the thickness direction (the direction of the arrow X) of the holding plate **34**. The holding plate **34** has a point-symmetric shape. The holding plate **34** includes a plate central portion **40** and a plurality of arm portions **42**.

[0039] The plate central portion **40** is a pressing portion **41** that presses the cell stack **12** in the stacking direction via the pressure receiving plate **36**. The plate central portion **40** is placed at a central portion of the holding plate **34**. The plate central portion **40** is located more inward in the stacking direction than an arm tip portion **44**, which is an end portion in the extending direction of the arm portion **42**. Therefore, when viewed from the direction perpendicular to the stacking direction, the holding plate **34** as a whole has a shape that is convex inward in the stacking direction. The plate central portion **40** is formed with a through hole **35**. The through hole **35** is formed at a central portion of the holding plate **34**. The through hole **35** is circular when viewed from the X direction. The function of the through hole **35** will be described later.

[0040] The arm portions **42** extend radially from the plate central portion **40**. The arm portions **42** are provided at equal intervals in the circumferential direction of the plate central portion **40**. The arm portion **42** is a leaf spring portion that is elastically deformed when a tightening load is applied to the cell stack **12**. In this embodiment, the holding plate **34** has four arm portions **42**. The number of arm portions **42** of the holding plate **34** may be three or less or five or more.

[0041] At one of the holding plates **34**, the arm tip portion **44** that is an end portion in the extending direction of the arm portion **42** is connected to one end portion of the connecting member **38** by, for example, welding or the like. At the other of the holding plates **34**, the arm tip portion **44** that is an end portion in the extending direction of the arm portion **42** is connected to the other end portion of the connecting member **38** by, for example, welding or the like. In other words, at the pair of holding plates **34**, the arm tip portions **44** are fixed (stuck) to the opposite ends of the connecting member **38** in such a way that positions thereof are not adjustable.

[0042] The arm tip portion **44** is located more outward than the cell stack **12** when viewed from the stacking direction (the direction of the arrow X) of the battery cells **18**. The arm tip portion **44** does not overlap with the terminal portions **22** when viewed from the direction of the arrow X. The arm tip portion **44** is provided with a positioning hole **34*a***. The function of the positioning holes **34*a*** will be described later.

[0043] The elastic force (spring force) of the four arm portions **42** is applied to the cell stack **12** as the tightening load via the pressure receiving plate **36**. This tightening load is the holding force of the holding plate **34** with respect to the cell stack **12**. The battery frame **16** holds the cell stack **12** solely by the frictional force generated by the force of the holding plate **34** pushing the cell stack **12** via the pressure receiving plate **36**.

[0044] The pressure receiving plate **36** is a pressing plate for evenly applying to the cell stack **12** the tightening load applied from the holding plate **34**. The pressure receiving plate **36** is placed between the holding plate **34** and the cell stack **12**. The pressure receiving plate **36** is formed in a square shape. The pressure receiving plate **36** is not essential. Therefore, the cell stack **12** may be directly pressed by the plate central portion **40** without the pressure receiving plate **36** being provided.

[0045] As shown in FIG. **3**, the pressure receiving plate **36** is formed with a threaded portion **37**. The function of the threaded portion **37** will be described later. Specifically, the threaded portion **37** is an internal thread. In this embodiment, the threaded portion **37** penetrates the pressure receiving plate **36**. The threaded portion **37** does not have to penetrate the pressure receiving plate **36**. The diameter of the threaded portion **37** is approximately the same as the diameter of the through hole **35** formed at the plate central portion **40** of the holding plate **34**. Specifically, the diameter of the threaded portion **37** is slightly smaller than the diameter of the through hole **35**. That is, the diameter of the through hole **35** is slightly larger than the diameter of the threaded portion **37**.

[0046] A first surface **36*a*** of the pressure receiving plate **36** facing the cell stack **12** is in surface contact with an end surface of the cell stack **12**. A second surface **36*b*** of the pressure receiving plate **36** facing in the direction opposite to the cell stack **12** is in surface contact with the plate center portion **40** of the holding plate **34**. The battery frame **16** may omit the pressure receiving plate **36**.

[0047] As shown in FIG. **1**, with the holding plate **34** being attached to the pressure receiving plate **36**, the four arm portions **42** extend, overlapping respectively with four corner portions of the pressure receiving plate **36** when viewed in the direction of the arrow X. With the holding plate **34** being attached to the pressure receiving plate **36**, a gap is provided between the arm portion **42** and the corner portions of the pressure receiving plate **36**. With the holding plate **34** being attached to the pressure receiving plate **36**, the four arm tip portions **44** are located more outward than the pressure receiving plate **36** when viewed in the direction of the arrow X.

[0048] The plurality of connecting members **38** connect the pair of holding plates **34** to each other in such a way that the tightening load (compressive load) is applied from the pair of holding plates **34** to the cell stack **12**. In this embodiment, the battery frame **16** has four connecting members **38**. Each connecting member **38** is a shaft that extends along the stacking direction of the battery cells **18**. The connecting member **38** is made of, for example, a metallic material, such as stainless steel.

[0049] The battery frame **16** configured as described above can move the pressing portion **41** in the direction to widen a space between the pair of holding plates **34** by the external force from the outside of the pair of holding plates **34**.

[0050] Next, a method of assembling the battery module **10** according to the first embodiment will be described.

[0051] The method of assembling the battery module **10** according to the first embodiment includes a region forming step, a placing step, and a pressing step.

[0052] As shown in FIGS. **4**A and **4**B, in the region forming step, a region R in which the cell stack **12** can be placed is formed between the pair of holding plates **34** by applying a coercive deformation force that deforms at least one of the pair of holding plates **34** against the elastic biasing force of the holding plates **34**. Although the case where both of the pair of holding plates **34** are deformed will be described here, only one of the holding plates **34** may be deformed.

[0053] The region forming step includes a preparatory step and a moving step. In the preparatory step, first, as shown in FIG. **4**A, an auxiliary jig **50** for coercive deformation is attached to the outside of the holding plate **34**. The auxiliary jig **50** has a leg portion **52** that is in contact with the arm tip portion **44** of the holding plate **34**. The leg portion **52** is provided with a positioning projection **53**. The positioning projection **53** is inserted into the positioning hole **34***a* provided at the holding plate **34**, whereby the auxiliary jig **50** is positioned with respect to the holding plate **34**. It is noted that the positioning holes **34***a* may be provided at the auxiliary jig **50** whereas the positioning projection **53** may be provided at the holding plate **34**.

[0054] In the preparatory step, the shaft portion **56***a* (external thread) of the bolt **56** is inserted into the hole portion **54** provided at a central portion of the auxiliary jig **50**. The bolt **56** is one aspect of a structural member for moving the plate center portion **40** in a direction to increase the space between the pair of holding plates **34**. The hole portion **54** is a hole in which no thread is formed. Thus, the shaft portion **56***a* of the bolt **56** cannot be screwed into the hole portion **54**. The shaft portion **56***a* of the bolt **56** penetrates the auxiliary jig **50** and is inserted into the through hole **35** provided in the holding plate **34**. At this time, the tip of the shaft portion **56***a* of the bolt **56** is slightly engaged with (screwed into) the threaded portion **37** provided at the pressure receiving plate **36**. Also, the head **56***b* of the bolt **56** is brought into contact with the auxiliary jig **50**.

[0055] As shown in FIG. **4**B, in the moving step of the region forming step, the bolt **56** is rotated while an axial position of the bolt **56** is fixed, whereby the plate central portion **40** is moved in the direction (outward in the X direction) in which the plate central portion **40** is brought closer to the auxiliary jig **50**. Specifically, when the bolt **56** inserted into the auxiliary jig **50** is rotated, the plate central portion **40** is displaced forcibly toward the auxiliary jig **50** against the elastic biasing force of the holding plate **34** by the screwing action between the shaft portion **56***a* of the bolt **56** and the threaded portion **37** of the pressure receiving plate **36**. The plate center portion **40** of each of the pair of holding plates **34** is displaced toward the auxiliary jig **50**, whereby a region R is formed in which the cell stack **12** can be placed between the pair of holding plates **34** (between the pair of pressure receiving plates **36** in the first embodiment).

[0056] Next, as shown in FIG. **5**A, in the placement step, the cell stack **12** is placed in the region R formed by the region formation step. It is preferable that the assembly method including the placement step is performed in such a way that the X direction is the vertical direction. In this way, the cell stack **12** can be placed on one of the pressure receiving plates **36** and thus the cell stack **12** can be easily placed in the region R formed by the region forming step. In FIGS. **5**A and **5**B, the cell stack **12** is shown in a simplified manner.

[0057] Next, as shown in FIG. **5**B, in the pressing step, in a state where the cell stack **12** is arranged in the region R, the above-described coercive deformation force is removed, whereby the cell stack **12** is pressed in the stacking direction by the elastic biasing force of the holding plate **34** by. Specifically, the bolt **56** is rotated in the direction opposite to the rotation direction of the bolt **56** in the region forming step and the bolt **56** is loosened, whereby the holding plate **34** and the pressure receiving plate **36** are displaced toward the cell stack **12**. Thereby, a tightening load is applied to the cell stack **12** by the pair of holding plates **34**.

[0058] The first embodiment provides the following benefits.

[0059] As shown in FIG. **5**A, by applying a coercive deformation force from the outside to the holding plate **34**, it is possible to form the region R in which the cell stack **12** can be placed. In addition, as shown in FIG. **5**B, by removing the coercive deformation force, it is possible to hold the cell stack **12** by the elastic biasing force of the holding plate **34**. This eliminates the need to tighten the holding plate **34** to the connecting member **38** by means of the screw structure, and the holding plate **34** and the connecting member **38** can be fixed in such a way that positions are not adjustable. Therefore, the stress is applied evenly to the plurality of connecting members **38**. As a result, the tightening force can be properly applied to the cell stack **12**.

[0060] As shown in FIG. **4**B, in the region forming step, a coercive deformation force is applied in a state where the end portion (arm tip portion **44**) in the extending direction of the arm portion **42** is fixed, whereby the arm portion **42** is deformed. This allows the arm portion **42** to be effectively deformed.

[0061] In the area forming step, the plate central portion **40** (pressing portion **41**) is moved in the direction opposite to the pressing direction in which the cell stack **12** is pressed in the pressing step. This makes it possible to properly form the region R in which the cell stack **12** can be placed.

[0062] In the region forming step, the bolt **56** is screwed into the thread portion **37** formed at the pressure receiving plate **36**, and the bolt **56** is rotated while the axial position of the bolt **56** is fixed, whereby the plate central portion **40** is moved. As a result, the plate central portion **40** can be moved effectively. The threaded portion **37** may be formed at the plate central portion **40**. In this case, in the region forming step, the bolt **56** is screwed into the threaded portion **37** formed at the plate central portion **40**, and the bolt **56** is rotated while the axial position of the bolt **56** is fixed, whereby the plate central portion **40** is moved. Alternatively, the threaded portion **37** may be formed at both the pressure receiving plate **36** and the plate center portion **40**.

[0063] In the region forming step, the bolt **56** inserted through the auxiliary jigs **50** attached to the holding plate **34** are rotated. This allows the plate center portion **40** of the holding plate **34** to be easily moved as the bolt **56** rotates.

[0064] Next, a method of assembling the battery module **10** according to the second embodiment will be described.

[0065] As shown in FIG. **6**, an engaging structure **70** that can engage with the displacement jig **60** is provided at the plate center portion **40** of the holding plate **34**. The engaging structure **70** has a central hole **72**, a slit **74**, and an engaging groove **76**. The slit **74** can receive the tip portion **62** of the displacement jig **60**. In FIG. **6**, a pair of slits **74** are formed extending from the central hole **72** in opposite directions. The engaging groove **76** can be engaged with the tip portion **62** of the displacement jig **60** inserted via the slit **74**. The engaging groove **76** communicates with the slit **74**.

[0066] The displacement jig **60** is one aspect of a structural member for moving the plate center portion **40** in a direction to increase the space between the pair of holding plates **34**. A pair of engaging pins **64** are provided at the tip portion **62** of the displacement jig **60**. The pair of engaging pins **64** are insertable into the pair of slits **74** of the engaging structure **70**. The pair of engaging pins **64** can be engaged with the engaging grooves **76** of the engaging structure **70**. A single engaging pin **64** and a single slit **74** may be provided. Three or more engaging pins **64** and three or more slits **74** may be provided.

[0067] In the method of assembling the battery module **10** according to the second embodiment, the region forming step includes an inserting step, an engaging step, and a moving step. In the inserting step, the tip portion **62** (the pair of engaging pins **64**) of the displacement jig **60** is inserted into the pair of slits **74** of the engaging structure **70**. Next, as shown in FIG. **7**A, in the engaging step, the displacement jig **60** is rotated, whereby the engaging pins **64** of the displacement jig **60** are engaged with the engaging grooves **76** of the engaging structure **70**.

[0068] Next, as shown in FIG. **7**B, in the moving step, the displacement jig **60** is pulled, whereby the holding plate **34** is forcibly displaced and the plate center portion **40** is moved to the outward direction of the battery frame **16** (outward in the X direction). Thus, a region R in which the cell stack **12** can be placed is formed between the pair of holding plates **34** (between the pair of pressure receiving plates **36** in the second embodiment).

[0069] Although not shown, the placing step and the pressing step are performed subsequently in the same manner as in the first embodiment. Specifically, in the second embodiment, the cell stack **12** is placed in the formed region R. Thereafter, the pulling force on the displacement jig **60** is released, whereby the coercive deformation force on the holding plate **34** is removed and the holding plate **34** and the pressure receiving plate **36** are displaced toward the cell stack **12**. Thereby, a tightening load is applied to the cell stack **12** by the pair of holding plates **34**.

[0070] The second embodiment also can properly apply the tightening force to the cell stack **12**. In the second embodiment, the displacement jig **60** is engaged with the engaging structure **70** and further the displacement jig **60** is pulled, whereby the holding plate **34** is deformed forcibly. Therefore, the region R where the cell stack **12** can be placed between the pair of holding plates **34** can be formed by a simple method. The engaging structure **70** may be provided on the pressure receiving plate **36**. In this case, the plate central portion **40** is provided with an insertion hole through which the displacement jig **60** is inserted.

[0071] Next, a method of assembling the battery module **10** according to the third embodiment will be described.

[0072] As shown in FIG. **8**A, a permanent magnet **80** is fixed to the plate center portion **40** of each of the pair of holding plates **34**. The region forming step in the assembling method according to the third embodiment includes a preparatory step and a moving step. In the preparatory step, a pair of electromagnets **82** are placed outside the battery frame **16** so as to face, with a space, the permanent magnets **80** fixed to the pair of holding plates **34**.

[0073] Next, as shown in FIG. **8**B, in the moving step, the plate central portion **40** is moved by external magnetic force. Specifically, the pair of electromagnets **82** are energized, whereby the permanent magnets **80** are pulled toward the electromagnets **82** by magnetic force. Thus, the plate central portion **40** is moved toward the outside of the battery frame **16** (outward in the X direction). Thus, a region R in which the cell stack **12** can be placed is formed between the pair of holding plates **34** (between the pair of pressure receiving plates **36** in the third embodiment).

[0074] Although not shown, the placing and pressing steps are then performed in the third embodiment as in the first embodiment. Specifically, in the third embodiment, the cell stack **12** is placed in the formed region R. Thereafter, energizing of the pair of electromagnets **82** is stopped, whereby the coercive deformation force on the holding plate **34** is removed and the holding plate **34** and the pressure receiving plate **36** are displaced toward the cell stack **12**. Thereby, a tightening load is applied to the cell stack **12** by the pair of holding plates **34**.

[0075] The third embodiment also can properly apply the clamping force to the cell stack **12**. In the third embodiment, the plate central portion **40** is moved by external magnetic force. Therefore, the region R where the cell stack **12** can be placed between the pair of holding plates **34** can be formed by a simple method. The plate central portion **40** may be configured in such a way that the permanent magnet **80** is attachable and detachable. When the plate central portion **40** is made of a metal material (ferromagnetic material) that can be attracted by the magnetic force of the permanent magnet **80**, the permanent magnet **80** may be omitted.

[0076] With respect to the above embodiments, we further disclose the following supplementary note.

(Supplementary Note **1**)

[0077] Concerning a method for assembling a battery module (**10**) of the present disclosure, the battery module includes: a cell stack (**12**) that includes battery cells (**18**) and heat exchangers (**20**) stacked on the battery cells; and a holding mechanism (**17**) that holds both ends in the stacking direction of the cell stack, the holding mechanism includes a pair of holding plates (**34**) that are elastically deformable, and a plurality of connecting members (**38**) that connect the pair of holding plates with each other, and the cell stack is pressed inward in the stacking direction by the elastic biasing force of the pair of holding plates, whereby the cell stack is held. The method includes a region forming step of forming a region (R) where the cell stack is placeable between the pair of holding plates by applying a coercive deformation force that deforms at least one of the pair of holding plates against the elastic biasing force, a placement step of placing the cell stack in the region formed in the region forming step, and a pressing step of pressing the cell stack in the stacking direction by the elastic biasing force, by removing the coercive deformation force in a state where the cell stack is placed in the region.

(Supplementary Note **2**)

[0078] In the method described in Supplementary note **1**, each of the pair of holding plates may include a pressing portion (**41**) and an arm portion (**42**) that extends from the pressing portion, and in the region forming step, the arm portion may be deformed by applying the coercive deformation force in a state where an end portion in an extending direction of the arm portion fixed.

(Supplementary Note **3**)

[0079] In the method according to Supplementary note **2**, in the region forming step, the pressing portion may be moved in a direction opposite to the pressing direction in which the cell stack is pressed in the pressing step.

(Supplementary Note **4**)

[0080] In the method described in Supplementary note **3**, at least one of the pressing portion and the pressure receiving plate (**36**) superposed on the pressing portion may be provided with a threaded portion (**37**), and in the region forming step, the pressing portion may be moved by screwing a bolt (**56**) into the threaded portion and rotating the bolt while fixing the axial position of the bolt.

(Supplementary Note **5**)

[0081] In the method according to Supplementary note **4**, in the region forming step, the bolt inserted through an auxiliary jig (**50**) attached to the holding plate may be rotated.

(Supplementary Note **6**)

[0082] In the method for assembling a battery module according to Supplementary note **3**, the pressing portion or the pressure receiving plate superposed on the pressing portion may be provided with an engaging structure (**70**) including a slit (**74**) and an engaging groove (**76**), the region forming step may include an inserting step of inserting a tip portion of a structural member into the slit of the engaging structure, an engaging step of engaging the tip portion with the engaging groove by rotating the structural member, and a moving step of moving the pressing portion by pulling the structural member.

(Supplementary Note **7**)

[0083] In the method according to Supplementary note **3**, in the region forming step, the pressing portion may be moved by an external magnetic force.

(Supplementary Note **8**)

[0084] A holding mechanism of the present disclosure includes a pair of holding plates that are elastically deformable face each other and a plurality of connecting members that connect the pair of holding plates with each other, wherein the holding mechanism is configured to hold a holding object placed between the pair of holding plates by elastic biasing force of the pair of holding plates, each of the plurality of holding plates includes a pressing portion and a plurality of arm portions extending from the pressing portion toward the plurality of connecting members, the plurality of arm portions are connected to the plurality of connecting members, and the pressing portion is configured to be moved in the direction to increase a space between the pair of holding plates by the external force from the outside of the pair of holding plates.

(Supplementary Note **9**)

[0085] In the holding mechanism according to Supplementary note **8**, a structural member for moving the pressing portion in the direction to increase the space may be attachable to the pressing portion.

(Supplementary Note **10**)

[0086] In the holding mechanism according to Supplementary note **9**, the pressing portion may be formed with a threaded portion configured to be engaged with the structural member.

(Supplementary Note **11**)

[0087] In the holding mechanism described in Supplementary note **9**, the pressure receiving plate superposed on the pressing portion may be formed with a threaded portion configured to be engaged with the structural member.

(Supplementary Note **12**)

[0088] In the holding mechanism according to Supplementary note **11**, the pressing portion may be formed with a through hole (**35**) through which the bolt, which is the structural member, is inserted.

(Supplementary Note **13**)

[0089] In the holding mechanism according to Supplementary note **12**, the diameter of the through hole may be approximately the same as the diameter of the threaded portion.

(Supplementary Note **14**)

[0090] In the holding mechanism according to Supplementary note **12** or **13**, the through hole may be formed at a central portion of the holding plate, and the threaded portion may be formed in a central portion of the pressure receiving plate.

(Supplementary Note **15**)

[0091] In the holding mechanism according to Supplementary note **9**, the pressing portion or the pressure receiving plate superposed on the pressing portion may be provided with an engaging structure configured to be engaged with the structural member, and the engaging structure may include a slit configured to receive a tip portion of the structural member and an engaging groove configured to be engaged with the tip portion inserted via the slit.

(Supplementary Note **16**)

[0092] In the holding mechanism according to Supplementary note **8**, a permanent magnet (**80**) for moving the pressing portion in the direction to increase the space may be fixed to or attachable to the pressing portion.

[0093] Although the present disclosure has been detailed, the present disclosure is not limited to the individual embodiments described above. These embodiments may be variously added, replaced, altered, partially deleted, etc., without departing from the scope of the present disclosure or the intent of the present disclosure as derived from the claims and their equivalents. These embodiments can also be implemented in combination. For example, in the above-described embodiment, the order of the operations and the order of the processes are shown as an example, and are not limited to these. The same applies to the case where numerical values or mathematical expressions are used in the description of the above-described embodiment.

1. A method for assembling a battery module,
wherein
the battery module includes:
a cell stack that includes battery cells and heat exchangers stacked on the battery cells, and
a holding mechanism that holds both stacking-direction ends of the cell stack,
the holding mechanism includes
a pair of holding plates that are elastically deformable, and
a plurality of connecting members that connect the pair of holding plates with each other, and
the cell stack is pressed inward in a stacking direction of the cell stack by elastic biasing force of the pair of holding plates, whereby the cell stack is held,

the method comprising:

forming a region where the cell stack is placeable between the pair of holding plates by applying a coercive deformation force that deforms at least one of the pair of holding plates against the elastic biasing force;

arranging the cell stack in the region formed in the forming; and

pressing the cell stack in the stacking direction by the elastic biasing force, by removing the coercive deformation force in a state where the cell stack is placed in the region.

2. The method for assembling the battery module according to claim **1**, wherein

each of the pair of holding plates includes a pressing portion and an arm portion that extends from the pressing portion, and

in the forming, the arm portion is deformed by applying the coercive deformation force in a state where an end portion in an extending direction of the arm portion is fixed.

3. The method for assembling the battery module according to claim **2**, wherein

in the forming, the pressing portion is moved in a direction opposite to a pressing direction in which the cell stack is pressed in the pressing.

4. The method for assembling the battery module according to claim **3**, wherein

at least one of the pressing portion or a pressure receiving plate superposed on the pressing portion is provided with a threaded portion, and

in the forming, the pressing portion is moved by screwing a bolt into the threaded portion and rotating the bolt while fixing an axial position of the bolt.

5. The method for assembling the battery module according to claim **4**, wherein

in the forming, the bolt inserted through an auxiliary jig attached to the holding plate is rotated.

6. The method for assembling the battery module according to claim **3**, wherein

the pressing portion or a pressure receiving plate superposed on the pressing portion is provided with an engaging structure including a slit and an engaging groove, and

the forming includes

inserting a tip portion of a structural member into the slit of the engaging structure,

engaging the tip portion with the engaging groove by rotating the structural member, and

moving the pressing portion by pulling the structural member.

7. The method for assembling the battery module according to claim **3**, wherein

in the forming, the pressing portion is moved by external magnetic force.

8. A holding mechanism comprising:

a pair of holding plates that are elastically deformable and face each other; and

a plurality of connecting members that connect the pair of holding plates with each other,

wherein

the holding mechanism is configured to hold a holding object placed between the pair of holding plates by elastic biasing force of the pair of holding plates,

each of the plurality of holding plates includes

a pressing portion, and

a plurality of arm portions extending from the pressing portion toward the plurality of connecting members, and

the plurality of arm portions are connected to the plurality of connecting members, and

the pressing portion is configured to be moved in a direction to increase a space between the pair of holding plates by external force from outside of the pair of holding plates.

9. The holding mechanism according to claim **8**, wherein

a structural member that moves the pressing portion in the direction to increase the space is attachable to the pressing portion.

10. The holding mechanism according to claim **9**, wherein

the pressing portion is formed with a threaded portion configured to be engaged with the structural member.

11. The holding mechanism according to claim **9**, wherein

a pressure receiving plate superposed on the pressing portion is formed with a threaded portion configured to be engaged with the structural member.

12. The holding mechanism according to claim **11**, wherein

the pressing portion is formed with a through hole through which the bolt, which is the structural member, is inserted.

13. The holding mechanism according to claim **12**, wherein

a diameter of the through hole is approximately equal to a diameter of the threaded portion.

14. The holding mechanism according to claim **12**, wherein

the through hole is formed at a central portion of the holding plate, and

the threaded portion is formed at a central portion of the pressure receiving plate.

15. The holding mechanism according to claim **9**, wherein

the pressing portion or a pressure receiving plate superposed on the pressing portion is provided with an engaging structure configured to be engaged with the structural member, and

the engaging structure includes

a slit configured to receive a tip portion of the structural member, and

an engaging groove configured to be engaged with the tip portion inserted through the slit.

16. The holding mechanism according to claim **8**, wherein

a permanent magnet for moving the pressing portion in the direction to increase the space is fixed to or attachable to the pressing portion.

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